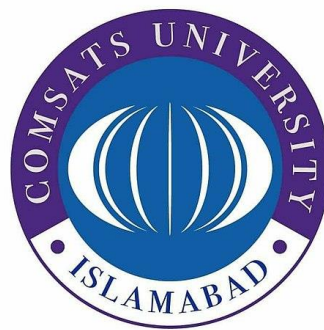


AI Powered University Management System

Final Year Project Proposal

Session 2023-2027

A project submitted in partial fulfilment of the requirements for the
Degree
of
BS in Computer Science / Software Engineering / Artificial Intelligence



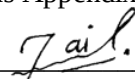
Department of Computer Science
COMSATS University Islamabad (CUI), Lahore Campus

Project Registration

Project ID (for office use)					
Type of project	☐ Traditional ☐ Industrial ☒ Continuing				
Nature of project	☒ Development ☐ Research & Development				
Sustainable Development Goals(SDGs)	☐ Good Health and Well-Being ☒ Quality Education ☒ Industry, Innovation, and Infrastructure ☐ Gender Equality ☐ Decent Work and Economic Growth ☐ Climate Action				
Area of specialization	☒ Artificial Intelligence (AI) ☐ Blockchain ☐ Cybersecurity ☐ Data Science and Analytics ☐ Game Development ☐ Internet of Things (IoT) ☒ Natural Language Processing (NLP) ☐ Mobile App Development ☒ Web Development				
Project Group Members					
Sr.#	Reg. #	Student Name	Email ID	Phone #	Signature
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Declaration: FYP group members have cleared all prerequisite courses For FYP-I as per their degree requirements. <u>For BS(Computer Science) and BS(Software Engineering)</u> (CSC241 Object Oriented Programming, CSC291 Software Engineering Concepts, CSC270 Database Systems, HUM102 Report Writing Skills) <u>For BS(Artificial Intelligence)</u> (CSC291 Software Engineering Concepts, HUM102 Report Writing Skills)					

Plagiarism Free Certificate

This is to certify that, I am Muhammad Zaid Amjad S/D/o Hafiz Muhammad Amjad, group leader of FYP under registration no CUI/SP23-BSE-135/LHR at the Computer Science Department, COMSATS University Islamabad, Lahore Campus. I declare that my FYP proposal is checked by my supervisor, and the similarity index is _____% that is less than 20%, an acceptable limit by HEC. The report is attached herewith as Appendix A.

Date: 03/10/2025 Name of Group Leader: Muhammad Zaid Amjad Signature: 

Name of Supervisor: Dr. Adeel Nawab Co-Supervisor (if any): _____

Designation: _____ Designation: _____

Signature: _____ Signature: _____

ABSTRACT

A University Management System is essential for efficiently managing academic, administrative, and financial operations by integrating all university processes into a single, streamlined digital platform. However, existing UMS lack the integration of state-of-the-art AI tools to further enhance and optimize their performance. This project aims to develop a UMS with three main AI tools: (1) CPGA / Subject Marks Predictions - to alert both students and teachers at the start of the semester to increase student performance and decrease dropout rate, (2) AskAI Agent - to facilitate students, teachers, and staff to ask questions in natural language to fulfil their information needs, and (3) Interactive Dashboards - provides reports with insights and predictions to enable students, teachers, and management to improve their decision making.

Previous Project Objectives and Features

Previous Project	Our AI Powered Project
Previous systems were traditionally built and offered static gateways for students, teachers, and admin.	The solution integrates AI-enabled dashboards, student performance prediction to bring data-driven learning and management more personalized.
They offered nothing more than computerization of a manual process, access to online forms, and maintaining records centrally in a database.	It provides smart predictions and insights to alert both students and teachers at the start of semester to increase student performance.
They didn't have predictive analysis of information, smart observations, or real-time analysis.	The solution uses the machine learning capabilities by training data from students marks and their feedback to the course and instructor, and provide AskAI agent gives important information whatever the user wants. i.e, student, teacher.
Their interfaces were very simple with limited features and not good to support decision making of teachers or learners..	The project transforms higher education systems from simple UI and functionality into intelligent and insight-based educational institution management with high quality analysis on dashboards.
Feedback and evaluation processes were done manually, students gave random and ineffective feedback and were sometimes inefficient, which didn't give enough value for much application of analytics to the growth of the institution and were almost useless.	The solution provides effective and secure data scalability and cloud deployment easily and effectively.

1. Introduction

The quick development of educational technology (EdTech) has changed the manner in which institutions used to manage overall learning and administration works. Higher education today are forced to adopt e-solutions that do more than just automate the overall learning process precisely and effortlessly. The standard way for the administration to be carried out has always been using

University Management Systems (UMS), which is mainly used for the management of course scheduling and staff scheduling. Still most of these tools are fixed, simple and don't include insights. For the revolutionized world of Technology, AI has reimagined our simple and basic university systems into something more flexible and AI-powered education systems. This solution can enhance student retention [1].

AI is rapidly growing in almost every field of life including Health, Finance, Automotive, etc. The capabilities of AI should be used in Higher Education to enhance the whole learning experience and predict valuable outcomes for the faculty backed by data. There is a huge gap in Higher Education as many current Institutes still not take advantage of AI. The old and current systems lack in modern requirements and useful features which includes feedback generation, predictive analysis for helping teachers and students, personalized learning experiences, etc. Hence, there is a universal need for integrating AI into UMS to provide wiser learning and teaching experience with modern standards.

Many university systems which have limited resources or just starting out, buy traditional software which create difficulties in providing modern education and market ready resources. These systems are not smart, personalized and not interactive and they do not enhance the learning outcomes. Current solutions are costly and do not use predictive analytics to measure the performance or feedback. According to recent studies, schools adopting AI driven tools see improved student retention and engagement. Therefore, the problem of unsmart and ineffective university systems needs to be addressed in order to embrace more productive learning environments and teaching methodologies. The main motivation to address this immense problem is to prepare our future youth according to the latest norms.

To overcome these problems, this project proposes an AI based University Management System which would be easy to deploy in any university. The system will be with four major portals which are designed for administrators, registration officers, staff, and students and ensure separation of concerns. AI features will include interactive dashboards, teacher and student performance predictive analytics, subject-wise or semester-wise, and auto-feedback generation tools for teachers based on reviews provided by students. It implements ML based AskAI Agent to facilitate stakeholders with important information to ask questions in natural language. It has a suite of tools that will empower students to track their own progress, help teachers identify which students may need additional support, and provide administrators with relevant data. This system is fit for any university providing a scalable, cost-effective and very easy deployment for them.

According to the modern era, the outcome will be a smarter and efficient university management system to provide higher quality of education. The students will be able to provide personalized information, and the teachers can choose their teacher based on student feedback. And the teachers will know students' achievements instead of just who was falling short. Whether planning and prioritizing resources or making decisions, managers can use real-time information. In the final analysis, it is expected that learning effectiveness will be enhanced and user satisfaction with the services improved, thus helping degree holders to more effectively transition into work. Success will be decided through usability testing, accuracy of AI prediction performance, and user feedback from pilot user evaluation.

2. Problem Statement

Nearly almost all of the University Management System (UMS) that currently exist are static and rule based. They automate administrative actions only, which lacks intelligent decision support. The old systems are static, giving personalized learning paths or predictive analysis only, and do not support faculty evaluation optimization or student performance improvements precisely and effectively. Until you add AI-powered feedback options, it creates waste of time, budget and poor educational evaluations.

Universities therefore struggle to manage with how to balance personal learning experiences with consistency for the efficiency of the institutes. To achieve this, there has always been a growing need for AI-UMS that uses predictive analytics, NLP and intelligent dashboard for learning through data and manage the whole Higher Education system.

3. Project Objectives

- To design an AI-driven University Management System which would include predictive analysis and NLP-based feedback analysis for the best of the students and teachers.
- To plan separate, role-specific portals with secure and reliable accessible control features for individuals. i.e. for Admin, Faculty, Staff, and Students.
- To develop highly interactive dashboards for the presentation of real-time performance data and institutional trends which would provide high quality insights and quick review for overall performances.
- To integrate AI models in the systems to predict students' performance and faculty performance on the basis of historical trends and feedback provided by the students.
- To provide secure, easily accessible, scalable, and cloud-ready deployment through modern DevOps practices for secure and reliable environment.

4. Related work

Ref #	Authors & Title	Type	Key Findings	Limitations
[2]	Smart Campus System - China	Journal	Integrates AI for attendance, performance tracking, and resources Enhances communication between students, teachers, and administration.	Implementation requires high initial cost and technical infrastructure. Data privacy and security concerns due to centralized information storage.
[3]	EduSys University Management (Global)	System	Platform managing academic and administrative functions. Cloud-based, offering dashboards and analytics.	Lacks advanced AI-based prediction or NLP-driven feedback systems.
[4]	Netkampuss AITS – AI-Enabled UMS (India)	System	Integrates AI for smart automation and predictive insights. Centralized management of departments, exams, and staff performance.	Limited details on specific AI models or algorithms used.
[5]	Enhancing personalized learning in campus	Journal (ScienceDirect)	Current UMS handle enrolment, scheduling, resources. AI integration	Limited empirical validation of proposed AI.

	management systems		enhances personalization.	Sample size and institutional context not clearly specified.
[6]	How AI & Predictive Analytics are Transforming Higher Education	Industry Report (Academia ERP)	AI saves up to 30% administrative costs. Reduces human error and improves data accuracy.	Details not fully transparent Cost savings claims may not generalize across all institution types and sizes.
[7]	Human-centred learning analytics and AI in education review"	Journal (ScienceDirect)	Excluding stakeholders from design leads to systems that don't meet needs. Privacy and agency concerns are critical.	Systematic review methodology not fully disclosed. Theoretical focus with fewer concrete implementation examples.
[8]	AI-Driven Evolution in Learning Analytics for Digital Education	Industry/Academic Report	LMS platforms incorporating AI features Examples include Moodle Analytics with performance prediction and automated reporting	Heavy focus on specific platforms (Moodle) Limited discussion of interoperability and standardization issues.

4.1 Relevance to Our Work and Addressed Gaps

The present AI learning offerings function in silo-based manners too, being able to do no more than one thing like prediction, dashboards and feedback and therefore cause fragmentation. The proposed UMS provides a solution to this challenge, since it allows the unification of several learning functionalities in an all-encompassing system. In addition, existing systems cater to only a generalized user group either students or instructors whereas this multi-portal approach is general since it provides customized interfaces for administrators, teachers and students. Immediately available UMS platforms are also old style and expensive, with no built-in AI capabilities for decision-making based on data. However, the incorporation of interactive dashboards and interpretable insights in the new system enhances trust and explainability. Additionally, there are few empirical studies relating to AI-based learning systems up to this time in low or medium-HDI nations.

5. Proposed Methodology and Architecture

This product is developed with weekly or biweekly sprint-based agile approach to ensure smooth and collaborative workflow within all the teams. We use collaboration tools like Jira, ClickUP or Trello to ensure transparency and effortless teamwork. The product first ensures all the current flows and functions of the University Management System having four dedicated portals. AI driven dashboards provide valuable information and insights used to predict the class-wise, subject-wise and individual student performances using NLP models trained on previous data of Students. AI powered Faculty evaluations using Feedback data of the institute helps students and organizations to improve performance and quality of Education. To ensure ease of use, accuracy, and performance, each function will be regularly examined and improved based on actual user input.

5.1 System Architecture

The AI-based University Management System is made up in different layers to be efficient, scalable, and intelligent as required by modern technology. Figure.1 shows the frontend layer has interactive dashboards and separate portals for students, teachers, administrators, and staff. Each portal offers respective users' role-specific features like monitoring performance, and data visualization. The overall processing and system logic for example, authentication, user management, AI analytics, and performance evaluation are managed in the application layer. It covers the UIs and the back end and applies business rules and predictive models to deliver insights. The data layer securely stores the academic records, user records, feedback, and AI model output to provide access control and privacy in a database. The AI and analytics layer perform predictive analysis, measurement of the feedback, and intelligent report generation to help in the identification of trends of student achievement and instructor's performance effectively and precisely.

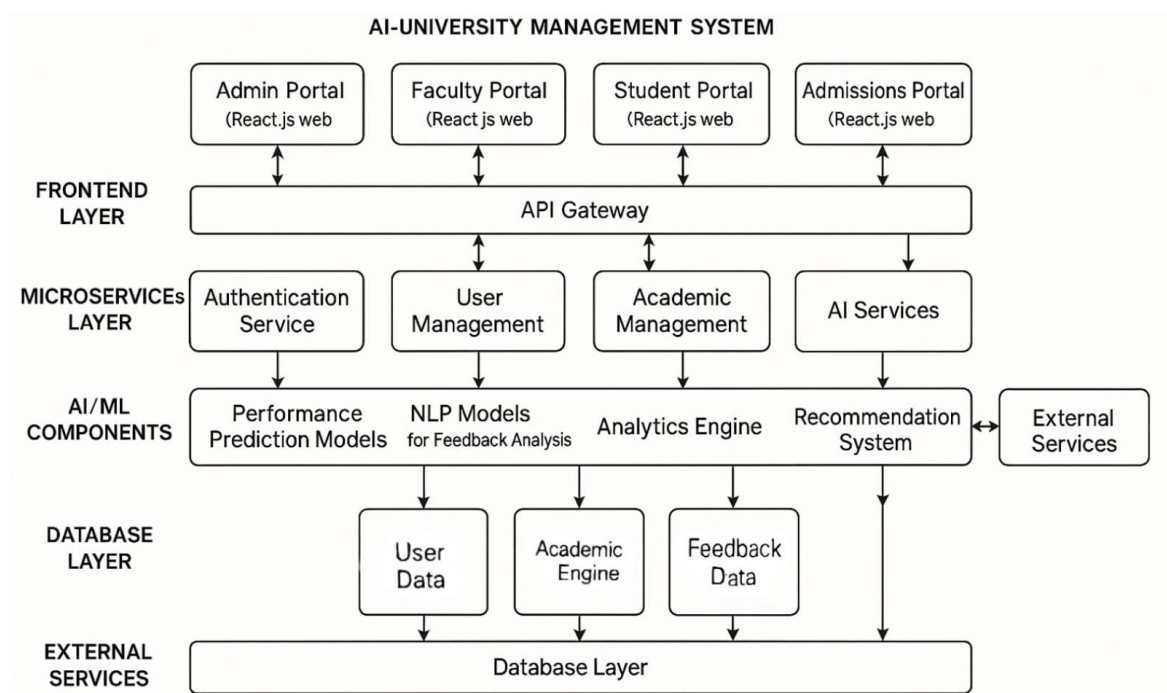


Figure 1. Architecture Diagram.

6. Success Criterion

The success of the project is based on the following of objectives and goals:

- **Adaptability for any University**

The product must be adaptable and customizable for any university. It provides institutional registration and after successful registration the specific University will use it.

- **Four Personalized Consoles/Portals with Role Based Access Control**

It must include four personalized consoles for every role within a University. This includes Student portal, Faculty portal, Staff portal and Admin portal.

- **Interactive AI powered Dashboards for Each Role**

AI Powered Dashboards for students and teachers provides insights and analytics. Also, the dashboards represent data in interactive graphs and charts. It provides reports with insights and predictions to enable students, teachers, and management to improve their decision making.

- **Smart AI Predictions for Students and Teachers**

This system should provide accurate and smart predictions for individual students based on the previous data, attendance, scores, etc. Also, multiple predictions of students for the teachers based on individual student, subject wise predictions of a specific student, classroom-based performance predictions and semester wise predictions.

- **ML based Model – ASK AI**

It must provide ML trained on previous institutional data, to facilitate students, teachers, and staff to ask questions in natural language to fulfil their information needs.

- **Elegant User Interface and User Experience**

This product must have a clean, sleek and easy to use User Interface build with ReactJs, TailwindCSS and NextJs. This combination will provide fast, efficient and modern Interface which significantly enhances the User Experience.

- **Data Security and Compliance**

All institutions data must be secure and separate. The system should ensure privacy of important data of institutions. It must comply with the security standards to responsibly handle all the data.

- **Secure and Scalable System Architecture**

The system should implement secure authentication using JWT, Cookies and OAuth 2.0. The system also save data in encrypted form to ensure privacy. Systems architecture must be scalable and reliable. System must be loosely coupled and high cohesion throughout the system.

- **Effortless Deployment**

The system should be deployed on AWS/Azure/GCP server. It should be containerized with Docker and Orchestrated with Kubernetes K8s. The system should be available and provides maximum uptime.

- **User Acceptance and Ease of Use**

All users Staff, Administrators, Students and Faculty should use the system easily and they must do all their respective tasks effortlessly.

- **Impact on Educational Outcomes**

This product should improve student engagement, student job readiness. Faculty should be able to better guide the students in which area they are weak. Positive feedback from faculty

should indicate that AI insights helped them adjust teaching strategies or provide targeted support.

7. Team Roles and Responsibilities

Our team roles and responsibilities are tentatively covered in the Table 1 below.

Table 1. Individual Tasks

Team Member	Activity	Tentative Date
All members	Requirements Gathering	October 2025 – November 2025
All members	Documentation & Diagrams	November 2025 – December 2025
Abdullah , Saad	Data Collection, Labelling & Preprocessing (NLP)	December 2025 – January 2026
Abdullah , Saad	Model Design & Training	January 2026 – February 2026
Abdullah , Zaid	Model Evaluation & Testing	February 2026 – March 2026
Zaid , Saad	Model Integration & deployment	March 2026 – April 2026
Zaid , Abdullah	Backend Development	April 2026 – June 2026
Zaid , Abdullah	API Integration	June 2026 – July 2026
Saad	Frontend Development	July 2026 – September 2026
All members	Manual & Automation Testing	September 2026 – October 2026
All members	Website & Service Deployment	October 2026 – November 2026

Zaid , Saad	Monitoring, Maintenance & Final Report	November 2026 – December 2026
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7.1 Gantt Chart

Figure 2 shows the tentative dates and tasks of the project in a graphical way using Gantt Chart. By using this diagram, we can clearly identifies the tasks divisions and tentative duration.

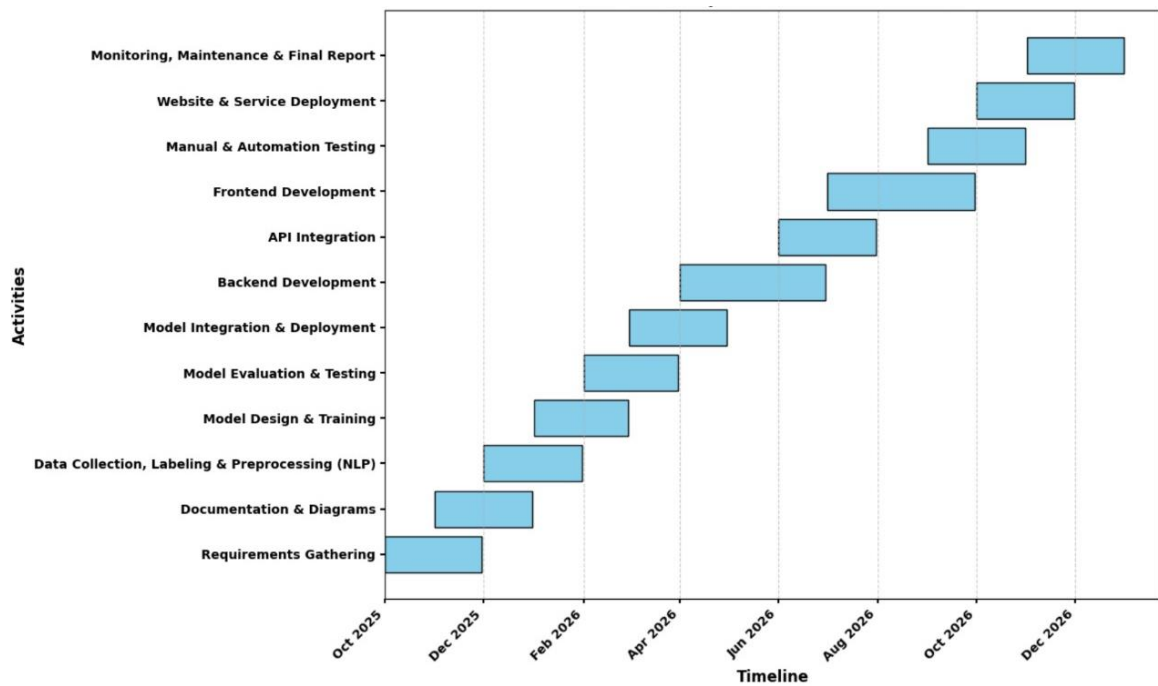


Figure 2. Gantt Chart.

8. Tools and Technologies

Design: Figma, Canva, Draw.io

Frontend: ReactJs, Redux UI, Shadcn, NextJs, TailwindCSS, JavaScript, TypeScript

Backend: Python, Django, FastAPI, PostgreSQL, MongoDB or AWS DynamoDB, DjangoORM and DjangoREST Framework for API development.

AI & ML Models: Scikit-Learn, TensorFlow, OpenAI or Gemini Models.

Project Management: Jira, Trello or ClickUp

Deployment and DevOps: AWS/Azure/GCP for cloud hosting, Prometheus and Grafana for system monitoring, Docker and Kubernetes for containerization and orchestration.

Version Control: Git and Github

Security and Encryption: OAuth2.0, JWT, Session Auth, HTTPS with SSL/TLS for secure communication, and AES-256 encryption for the storage of important information.

9. References

- [1] HSO. Study to make AI powered Higher Education for better student engagement. [Online]. Available: <https://www.hso.com/blog/enhancing-student-retention-with-ai-in-higher-education>. [Accessed 5 10 2025].
- [2] Smart Campus System, “AI-Driven Smart Campus Solutions for Universities in China,” *China Education Technology Journal*, vol. 45, no. 3, pp. 112–118, 2023.
- [3] EduSys, “University Management ERP Software – Cloud-based Solution for Education Institutions,” EduSys Official Website, 2024. [Online]. Available: <https://www.edusys.co/en-in/university-management-software-erp.html>
- [4] Netkampus, “AI-Enabled University Management System (AITS),” Netkampus Technologies Pvt. Ltd., 2023. [Online]. Available: <https://www.netkampus.com>
- [5] AI student success predictor: Enhancing personalized learning in campus management systems," *ScienceDirect Journal*, May 2024. [Online]. Available: <https://www.sciencedirect.com>
- [6] Modern Student Information Systems: How AI & Predictive Analytics are Transforming Higher Ed," *Academia ERP Industry Report*, Feb. 2025.
- [7] Human-centred learning analytics and AI in education: A systematic literature review," *ScienceDirect Computers & Education*, vol. 237, Mar. 2024. [Online]. Available: <https://www.sciencedirect.com>
- [8] AI-Driven Evolution in Learning Analytics for Digital Education," *Digital Learning Institute Research Report*, 2024.

Appendix A

FYP Proposal

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